

SRIJAN GUPTA

+1 608-598-7333 sg7345@nyu.edu LinkedIn Portfolio

Education

University of Wisconsin-Madison (Dec 2025) New York University (Est. Dec 2027)

B.S. in Computer Science, Data Science, Economics and M.S. in Computer Science

NYC, New York

Relevant Coursework

- Algorithms
- Machine Learning
- Distributed Systems
- Operating Systems
- Data Structures
- Artificial Intelligence
- Database Systems
- Software Engineering

Technical Skills

Languages: Python, Java, C++, Go, JavaScript/TypeScript, SQL (OOP, concurrency, async)

Frameworks & ML: FastAPI, Node.js, Express, Next.js, React, PyTorch, TensorFlow, scikit-learn, LangChain, RAG

Data & Retrieval: PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB, pgvector; hybrid search, SQL, , chunking

DevOps & Cloud: Docker, Kubernetes, AWS (EC2, S3, Lambda, ECS, RDS), GCP (Vertex AI, GKE, Cloud Functions), CI/CD (GitHub Actions, Jenkins), Terraform

Tools: Git, JIRA, Linux, Bash, pytest, Jest, OAuth 2.0, model evaluation, integration testing, debugging

Experience

SpecMe

Nov 2025 – Present

Co-Founder

Madison, WI (Hybrid)

- Built an **AI specification engine** with an **NLP parsing layer** and **LLM prompt pipelines** with structured output validation, ingesting unstructured input from Slack, GitHub, Jira, and transcripts.
- Architected an **event-driven backend** using **Next.js serverless functions**, **PostgreSQL**, **OAuth 2.0/SSO**, and **Stripe billing**, with webhook-based real-time sync across connected platforms.
- Executed a full **SSR/SSG migration** and **GitHub API** integration for PR-level spec delivery; shipped MVP to design partners while iterating on model output quality from pilot feedback.

Institute on Aging (UW-Madison)

Feb 2024 – Present

Student IT Support Specialist

Madison, WI

- Rebuilt backend infrastructure for **200+** researchers, improving performance by **35%** through **REST API** redesign, **PostgreSQL** query optimization, and **Redis** caching.
- Developed automated **ETL pipelines** processing **100K+** survey responses, reducing manual processing by **90%** through **Python** orchestration and robust data validation.
- Implemented production security and **CI/CD** via **OAuth2**, **JWT**, and **GitHub Actions**, ensuring **zero-downtime** blue-green deployments and **Pydantic** validation.

Digivate Labs

Jun 2024 – Aug 2024

Data Processing Engineer Intern

Remote

- Built a production **ML document-processing system** improving accuracy **34%** (F1: 0.68 to 0.91) and cutting latency **40%** (850 ms to 510 ms) via **semantic caching** and pipeline refactoring.
- Architected **GCP ML infrastructure (Vertex AI, GKE)** achieving **2.1x** throughput for **500K+** daily documents using **distributed computing** and dead letter queues.

Projects

ClearClause - Production RAG System | Python, FastAPI, LangChain, pgvector, K8s

- Developed a production **RAG system** achieving **0.87 precision@10** via hybrid search (dense embeddings + lexical BM25); reduced query latency **93%** (4.3s to 300ms) through **Redis caching** and async streaming.
- Orchestrated infrastructure on **Kubernetes** with HPA and blue-green deployments; implemented **PII redaction**, chunking strategies, and secrets management to meet legal data security standards.

RAG Evaluation & Optimization Framework | Python, embeddings, vector search

- Benchmarked **36 retriever configurations** across **HotpotQA** and **LegalBench-RAG**, mapping latency-cost-quality **Pareto frontiers** to guide retriever, reranker, and embedding model selection.
- Built a reproducible **evaluation harness** automating retrieval metrics (recall@k, precision@k, MRR), token-cost accounting, and latency profiling for systematic **model comparison**.

BuckyConnect - Real-Time Collaboration Platform | React, Node.js, GraphQL, AWS

- Architected a full-stack SaaS for **280+ concurrent users** with real-time editing and video chat, achieving **99.5% uptime** and **<500 ms latency** via Node.js microservices and **WebSocket/Redis Pub/Sub**.
- Deployed on **AWS ECS** with auto-scaling and CloudFront; optimized frontend using **Web Workers** and lazy loading, cutting bundle size **40%** and achieving **<500 ms time-to-interactive**.

Fraud Detection API | Python, FastAPI, scikit-learn, Redis, Kubernetes

- Architected a high-throughput **ML inference API** handling **500+ QPS** with **<100 ms p95 latency** using FastAPI async endpoints, **Redis** feature caching, and circuit breaker patterns for fault tolerance.
- Built an observability stack with **Prometheus**, **Grafana**, and **Jaeger** for distributed tracing; established a **CI/CD pipeline** via GitHub Actions with **Trivy** scanning and **90%+** test coverage.